

SparkFlow

SparkFlow bridges the Sparkplug B stream Host decodes onto a NATS/JetStream backbone, so every point stays live and available company-wide, with no tag behind it, past the point where a tag-counted system stops scaling.

Version 3.0.0Ignition 8.1.19+ and 8.3Runs on: the Host gateway contents

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Overview

1. Overview

SparkFlow publishes every point Host decodes onto a NATS JetStream stream as a named, timestamped, quality-stamped message, independent of whether a tag exists for it. Dashboards, historians, analysts and AI pipelines subscribe directly; the plant is asked for nothing extra. A key-value store holds the current value and quality of every point, so a new subscriber gets the last known value of everything the moment it connects, and the stream keeps history so a window of past data can be replayed. SparkFlow also carries a command bridge: a NATS request/reply call can drive a confirmed NCMD/DCMD write back to the field.

SparkFlow is optional and is not part of the core SparkBridge install. It runs on the Host gateway, on its own perpetual per-gateway license, and is installed after Provider, Host and Edge are running.

Requirements

2. What You Need

Ignition	8.1.19 or newer on the 8.1 line, or 8.3 (separate build). A gateway that can be restarted at will.
Java	Compiles against the Ignition 8.1.0 API, targets Java 11 bytecode.
Gateway role	The Host gateway. SparkFlow is not installed on an Edge-only gateway.
Network	A reachable <code>nats-server</code> with JetStream enabled, default <code>nats://127.0.0.1:4222</code> .
Other SparkBridge modules required	Provider, Host and Edge running first. SparkFlow reads from Host's decoded stream.

Install

3. Install

- 1. Download the release bundle for your Ignition line from the [download page](#), including SparkBridge-Flow.mod1 (optional add-on, bought separately from the core bundle).
- 2. Install after Provider, Host and Edge are running: Config > Modules > Install or Upgrade a Module, upload SparkBridge-Flow.mod1.
- 3. Accept the certificate and license once; confirm Config > Modules shows SparkFlow state = Running.

See docs/FLOW_MODULE.md in the bundle for the full configuration and lifecycle reference.

License

4. License

File	Per-gateway license key at <data>/sparkbridge-license.key. SparkFlow carries its own license, separate from the core suite key, at \$995 per Host gateway.
Re-check interval	SparkFlow re-checks the key file every 30 seconds. Installing, replacing or removing a key takes effect with no restart.

From release 2.7.0, SparkFlow runs on Ignition's standard two-hour module trial (the gateway's Reset Trial button restarts the clock). When the trial expires with no verified key naming the product, SparkFlow stops its dispatcher. A key dropped in after expiry lifts the pause on its own within the 30-second re-check, no button press and no restart. A verified key overrides the trial entirely.

Configure

5. Configure

All keys are -D system properties, read at startup as bootstrap/defaults. The tagfeed filter additionally hot-reloads from the sparkflow-config NATS KV bucket at runtime; nothing else currently hot-reloads.

Setting	Default	Meaning
sparkbridge.flow.enabled	true	Master on/off switch. When false, startup logs and returns without connecting to NATS or the broker.
sparkbridge.flow.nats.url	nats://127.0.0.1:4222	NATS core connect URL.
sparkbridge.flow.tagfeed	empty	Comma-separated NATS-subject-style filter list (* = one token, > = remainder) selecting the subset projected onto the managed tag provider. Empty means nothing projects. Hot-reloads from the sparkflow-config KV bucket.
sparkbridge.flow.allowWildcardTagfeed	false	Must be explicitly set to true before a tagfeed filter list may contain the whole-estate wildcard uns.>.
sparkbridge.flow.subjectprefix	uns.	Telemetry subject prefix. Present for future configurability; the shipped

Setting	Default	Meaning
<code>sparkbridge.flow.kvbucket</code>	<code>uns-last</code>	subject mapper currently hardcodes <code>uns.</code> , so changing this key alone has no effect on the wire format. KV bucket holding tag-less last-known-value and lifecycle state. Created automatically if absent.
<code>sparkbridge.flow.configbucket</code>	<code>sparkflow-config</code>	KV bucket SparkFlow watches for its own hot-reloadable config. Not created automatically; must be provisioned ahead of time. If absent at startup, SparkFlow logs an error and continues on the static bootstrap tagfeed filter with hot-reload disabled.
<code>sparkbridge.flow.twin.kvbucket</code>	<code>uns-model</code>	Digital-twin KV bucket. Created automatically if absent, unlike the config bucket.
<code>sparkbridge.flow.commandTimeoutMs</code>	<code>5000</code>	Timeout for the reverse (NATS request/reply to NCMD/DCMD) command path.
<code>sparkbridge.flow.commandWorkers</code>	<code>8</code>	Threads in the command-bridge worker pool.
<code>sparkbridge.flow.commandQueue</code>	<code>64</code>	Bounded queue in front of the command workers. When full, a new command is answered at once with <code>{"busy":true}</code> rather than queued.
<code>sparkbridge.flow.commandPerEdge</code>	<code>2</code>	Commands allowed in flight per group/edge. A dead edge pins at most this many workers.
<code>sparkbridge.flow.nats.credentialsFile</code>	<code>none</code>	NATS credentials file path, resolved through the secrets file in preference to a bare <code>-D</code> value.
<code>sparkbridge.flow.nats.tls.enabled</code>	<code>false</code>	Whether the NATS connection uses TLS.
<code>sparkbridge.flow.nats.tls.certPath / keyPath / caPath</code>	<code>none</code>	TLS cert/key/CA paths, resolved through the secrets file the same way as the credentials file.
<code>sparkbridge.flow.cmd.keysDir</code>	<code>empty</code>	Directory of signed-command verification keys (public keys only, not routed through the secrets file).
<code>sparkbridge.flow.cmd.requireSignature</code>	<code>false</code>	When true, an unsigned command request is rejected instead of falling back to the unsigned path.

Every subject, command subject and KV key includes a mandatory device slot: `uns.<group>.<edge>.<device|_>.<metric segment>[...]`, with a literal underscore meaning no device. Metric segment characters outside `[A-Za-z0-9_-]` are percent-style escaped as `=xx` (two hex digits); a literal `=` becomes `=3D`. This escaped form is canonical on the wire subject, the KV key, and the historian node key.

Operate

6. Operate

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. Watch for repeated `{"busy":true}` command responses, which indicate the command queue or per-edge limit is saturated rather than a dead command bridge.

On a NATS link loss, SparkFlow's dispatcher stops publishing until reconnected; it does not buffer independently of what NATS/JetStream itself retains. On restart, SparkFlow reconnects to NATS and re-subscribes to Host's decoded stream; the tagfeed filter reloads from the config bucket if present.

A metric segment that is permanently unmappable for an unsupported wire value type or a rejected group/edge/device identifier publishes one event to `uns.$events.reject` and is never queued for retry.

Verify

7. Verify

Confirm the Edge's Sparkplug traffic reaches NATS subjects under the `uns.` prefix, for example with a NATS CLI subscription to `uns.>`. See `docs/FLOW_MODULE.md` in the bundle for the subject naming contract and the KV bucket layout used for last-known values.

Troubleshoot

8. Troubleshoot

Symptom	Likely cause	Fix
No subjects appear under <code>uns.</code>	<code>sparkbridge.flow.enabled</code> is false, or tagfeed is empty	Confirm the module is enabled and set a non-empty tagfeed filter
Startup logs an error naming the config bucket	<code>sparkflow-config</code> KV bucket was never provisioned	Provision the bucket ahead of time; SparkFlow will not create it itself
Command calls return <code>{"busy":true}</code> repeatedly	Command queue full or per-edge limit reached on a dead/slow edge	Investigate the named edge; raise <code>commandWorkers/commandQueue</code> only after confirming the edge itself is healthy
Metric with spaces or punctuation in its name fails to publish	Pre-escaping defect in an old build, or an unsupported wire value type	Confirm the module escapes metric segments as <code>=xx</code> ; an unsupported value type instead routes to <code>uns.\$events.reject</code>
Dispatcher stops publishing with no error	Module trial expired, no verified key	Drop a valid <code>sparkbridge-license.key</code> in the data directory

9. Reference

License file	<code><data>/sparkbridge-license.key</code>
Module file	<code>SparkBridge-Flow.modl</code>
Default NATS URL	<code>nats://127.0.0.1:4222</code>
Last-value KV bucket	<code>uns-last</code>
Config KV bucket	<code>sparkflow-config</code> (not auto-created)
Digital-twin KV bucket	<code>uns-model</code>

Reject event subject `uns.$events.reject`

License re-check 30 seconds

Support

10. Support

Full install and config detail: [SparkBridge Docs, Flow](#). Product page: [SparkFlow](#). Questions: [Contact](#). Release history: [GitHub releases](#).